TQUK Functional Skills Qualification in Maths at Level 2

Examination Past Paper 9

Please complete the details below using black or blue ink. Use BLOCK CAPITALS.

Learner Name: _____

Learner Number: _____

Date: _____

Centre Name: _____

Instructions:

- read each question **carefully**
- answer **all** questions
- write your answers **clearly** in the spaces provided
- **check** your answers.

Information:

- this examination has **two** sections. These are clearly labelled
- you are **not allowed** to use a calculator for Section A
- you **are allowed** to use a calculator for Section B
- the **maximum** mark for this examination is 60
- the marks available for each question are shown in **bold** beneath each question.

OFFICIAL MFS MODEL SOLUTION · BENCHMARK EXEMPLARY PASS

60 / 60 Marks (100% Pass · Full Marks)

· Assessment: Past Paper 9 (Non-Calculator & Calculator)

· Complete stepped calculations, methods & mark scheme codes [MI/AI]

· Exam Board: TQUK (Training Qualifications UK · Ofqual Regulated)

· Pass standard: 34 / 60 marks (Achieved: 60 / 60 marks)

Verified by Tutor Abdul (Senior Assessor) · MasterFunctionalSkills.co.uk

Total marks	60		

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Section A: Non-calculator



There are **15 marks** available in this section.
You must **not** use a calculator in this section.
You will have **30 minutes** to complete this section.

1. Work out 0.082 + 0.49

Step 1: Align decimal places with padding zero:

0.082 + 0.490 = 0.572

Answer	0.572 ✓ *[1 mark]*

[1]

2. Work out 8.246 ÷ 4

Step 1: Short division by 4:

8 ÷ 4 = 2, 2 ÷ 4 = 0 r2, 24 ÷ 4 = 6, 6 ÷ 4 = 1 r2, 20 ÷ 4 = 5

Result = 2.0615

Answer	2.06 ✓ *[1 mark]*

[1]

Past paper

Questions continue on the following page

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3. Work out $4^2 - 10 \times 2$

Step 1: BIDMAS: Indices, then Multiplication, then Subtraction:

Indices: $4^2 = 16$

Multiplication: $10 \times 2 = 20$

Subtraction: $16 - 20 = -4$

Answer	-4 ✓ *[2 marks]*

[2]

4. Use rounding to check the calculation below:

$7.1423 \times 6.8967 = 49.25830041$

Show your working.

Step 1: Round each number to 1 significant figure:

$7.1423 \approx 7$ and $6.8967 \approx 7$ and $49.2583... \approx 49$

Step 2: Check multiplication using rounded values:

$7 \times 7 = 49$ (or $49 \div 7 = 7$)

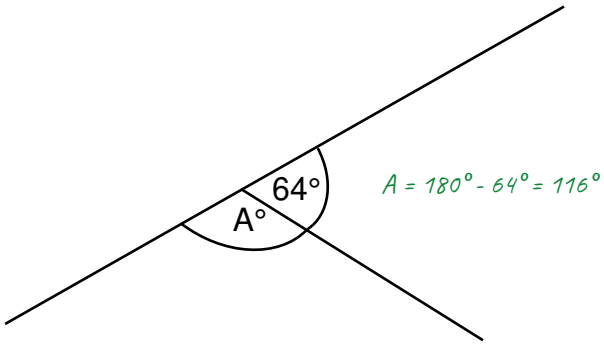
Answer	$7 \times 7 =$ ✓ *[2 marks]*

[2]

Past paper

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5. Casey thinks angle A is less than 120°:



A = 180° - 64° = 116°

Not drawn to scale

Is Casey correct?

Show how you decide.

Step 1: Angles on a straight line add to 180°:

Angle A = 180° – 64° = 116°

Step 2: Compare: 116° < 120°. Casey thinks it is less than 120°.

Therefore Casey is correct: Yes.

Answer	Yes ✓ [2 marks]

[2]

6. Work out:

738 925 – 245 128 – 18 503

Step 1: Subtract first number: 738 925 - 245 128 = 493 797

Step 2: Subtract second number: 493 797 – 18 503 = 475 294

Answer	475 2 ✓ [2 marks]

[2]

Questions continue on the following page

Past paper

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7. On a map, the distance between Mira's house and the cinema is 7 centimetres (cm).

The scale on the map is 1 : 50 000

Work out the actual distance between Mira's house and the cinema.

Give your answer in kilometres (km).

Step 1: Scale is 1 : 50 000. Measured distance = 7 cm
7 cm × 50 000 = 350 000 cm = 3.5 km

Actual distance = 7 cm × 50 000 = 350 000 cm

Step 2: Convert to kilometres (1 km = 100 000 cm):

350 000 ÷ 100 000 = 3.5 km

Answer	3.5 ✓ [2 marks]	**km**

[2]

8. The table shows the number of students on a course at college:

	First-year students	**Second-year students**
Attend online classes only	10	13
Attend classes on campus only	15	12

One person is chosen at random.

What is the probability that this person is a second-year student who attends classes on campus only?

Give your answer as a decimal **and** a percentage.

Step 1: Total = 10 + 13 + 15 + 12 = 50. 2nd-year campus = 12

Probability = 12 ÷ 50 = 0.24 = 24%

Answer	**a) Decimal:** 0.24	**b) Percentage:** 24% ✓ [3 marks]

[3]

End of Section A.

Past paper

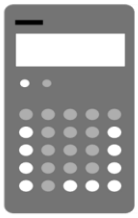
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Section B begins on Page 6.

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Section B: Calculator



There are **45 marks** available in this section.
You **can** use a basic calculator in this section.
You will have **90 minutes** to complete this section.

1. Write 12 540 000 in words.

Answer	Twelve million five hundred and forty thousand ✓ [1 mark]

[1]

2. Round 823.95132 to one decimal place.

Answer	824.0 ✓ [1 mark]

[1]

3. Bo asks a company to build a wall in 18 days.

The company's builders all work at the same speed.

It will take 5 builders 12 days to build the wall.

Can 3 builders build the wall in 18 days?

Show how you decide.

Step 1: Total workload = 5 workers × 12 days = 60 person-days

Step 2: Time for 3 workers = 60 ÷ 3 = 20 days

Step 3: Compare with 18 days limit: 20 > 18 days. Therefore No.

Answer	No ✓ [2 marks]

[2]

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4. Write 0.472:

a) as a fraction

b) as a percentage.

Step 1: 0.472 = 472 / 1000 = 236 / 500 = 59 / 125

Step 2: Percentage = 0.472 × 100% = 47.2%

Answer	a) Fraction: 59 / 125	b) Percentage: 47.2% ✓

[2 marks]

[2]

5. Put these fractions in order starting with the lowest:

$\frac{4}{5}$ $\frac{8}{5}$ $\frac{13}{15}$ $\frac{15}{10}$

Show any working here:

Step 1: Convert to common denominator 30:

4/5 = 24/30, 13/15 = 26/30, 15/10 = 45/30, 8/5 = 48/30

Answer	4 / 5	13 / 15	15 / 10	8 / 5 ✓
	Lowest	→		Highest

[2 marks]

[2]

Past paper

Questions continue on the following page

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6. Kai looks online at the reviews of a hotel.

1088 reviewers would recommend the hotel.

192 reviewers would **not** recommend the hotel.

What percentage of the total reviewers would recommend the hotel.

Step 1: Total responses = 1088 + 192 = 1280

Step 2: Percentage = (1088 ÷ 1280) × 100% = 0.85 × 100% = 85%

Answer	85 ✓ [2 marks]	%

[2]

Past paper

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7. Ashley buys a new freezer.

The instructions recommend that the freezer temperature is set at below 0 degrees Fahrenheit (°F).

Ashley sets the temperature to –15 degrees Celsius (°C).

Has Ashley set the freezer to the correct temperature?

Show how you decide.

Use:

$$C = \frac{5(F - 32)}{9}$$

Where:
C = temperature in degrees Celsius
F = temperature in degrees Fahrenheit

Step 1: Formula: C = 5(F - 32) ÷ 9

Substitute F = 0°F: C = 5(0 - 32) ÷ 9 = -160 ÷ 9 ≈ -17.8°C

Step 2: Compare: -15°C is warmer than -17.8°C (not as cold).

Therefore No.

Answer	**No** ✓ [3 marks]

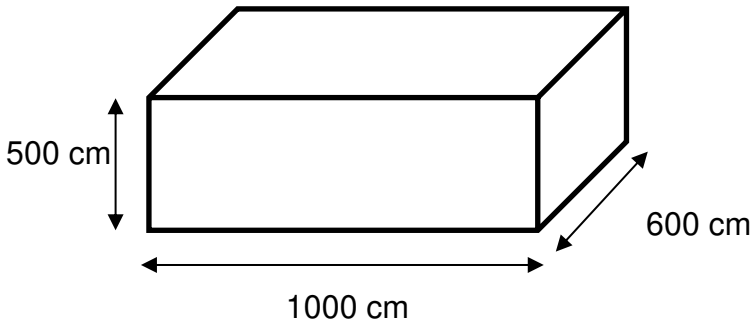
[3]

Past paper

Questions continue on the following page

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8. Charlie needs to paint the outside of this cuboid:



Not drawn to scale

Surface Area = 2(lh + lw + hw)
= 2 800 000 cm²

Charlie thinks the total area to be painted is more than 2 750 500 square centimetres (cm^2).

Is Charlie correct?

Show how you decide.

Step 1: Total Surface Area = 2 × (lh + lw + hw)
= 2 × (1000×500 + 1000×600 + 600×500)
= 2 × (500 000 + 600 000 + 300 000) = 2 × 1 400 000
= 2 800 000 cm²

Step 2: Compare: 2 800 000 > 2 750 500. Charlie is correct: Yes.

Answer Yes ✓ [3 marks]

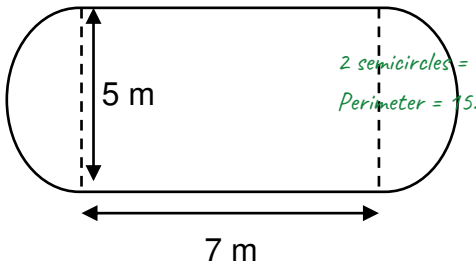
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9. Stevie needs to put a rope around the pond shown below.

The pond is in the shape of a rectangle with a semi-circle at each end.



2 semicircles = 1 circle: pi × 5 = 15.7 m
Perimeter = 15.7 + 7 + 7 = 29.7 m

Not drawn to scale

Stevie thinks 55 metres (m) of rope will be needed in total.

Is Stevie correct?

Show how you decide.

Use π = 3.14

Step 1: Perimeter = Semicircle arc + straight sides:

Two semicircles form 1 circle: pi × d = 3.14 × 5 = 15.7 m

Straight edges = 7 m + 7 m = 14 m --> Total = 15.7 + 14 = 29.7 m

Step 2: Compare: 29.7 m ≠ 55 m. Therefore Stevie is not correct (No).

Answer	No ✓ [3 marks]

[3]

Past paper

Questions continue on the following page

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10. Robin needs to make a pipe out of liquid copper.

Robin knows:

- density of copper = 8.96 g/cm^3
- the pipes need to have mass = 4480 grams.

What volume of liquid copper does Robin need to use?

Use: Density = $\frac{\text{mass}}{\text{volume}}$

Step 1: Rearrange formula: Volume = Mass ÷ Density

Mass = 4480 grams, Density = 8.96 g/cm³

Step 2: Calculate volume:

Volume = 4480 ÷ 8.96 = 500 cm³

Answer	500 ✓ *[3 marks]*	**cm^3**

[3]

Past paper

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11. Morgan checks how much driving instructors charge for driving lessons:

Amount (A) charged per hour (£)	Number of driving instructors
20 < A ≤ 30	2
30 < A ≤ 40	9
40 < A ≤ 50	3
50 < A ≤ 60	1
Total	15

Morgan estimates the mean amount per hour.

Morgan will not pay any more than the mean amount.

What is the maximum amount per hour that Morgan will pay?

Step 1: Midpoints and fx: 25×2=50, 35×9=315, 45×3=135, 55×1=55

Step 2: Total fx = 50 + 315 + 135 + 55 = 555

Step 3: Total frequency = 15 driving instructors

Step 4: Estimated Mean = 555 ÷ 15 = £37 per hour

Answer	£ 37 ✓ [4 marks]

[4]

Questions continue on the following page

Past paper

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12. Taylor is planning a road trip.

The total distance is 1860 miles.

The average speed limit is 30 miles per hour.

Taylor wants to drive for a maximum of 6 hours per day.

How many whole days will Taylor need to complete the **entire** road trip driving at the average speed limit?

Step 1: Total driving hours = 1860 miles ÷ 30 mph = 62 hours

Step 2: Maximum 6 hours per day:

Days needed = 62 ÷ 6 = 10.333... days

Step 3: Round up to nearest whole day: 11 days

Answer	11 ✓ [4 marks]	**days**

[4]

Past paper

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13. Izumi sells chocolate muffins and jam doughnuts.

This table shows the number of chocolate muffins sold over the last 6 days:

Chocolate muffins					
43	30	24	32	28	35

This table shows the number of jam doughnuts sold over the last 6 days:

Jam doughnuts	
Mean number sold per day	28
Range	25

Izumi claims:

i) 'on average I sell more chocolate muffins each day'
ii) 'the sales of the chocolate muffins are more consistent.'

Are both of Izumi's claims correct?

Give a reason for each of your answers.

Show your working.

Step 1: Chocolate Mean = (43+30+24+32+28+35) ÷ 6 = 192 ÷ 6 = 32

Step 2: Chocolate Range = 43 - 24 = 19

Reason - claim i)

Claim i is correct: Chocolate mean (32) > Jam doughnut mean (28).

Reason - claim ii)

Claim ii is correct: Chocolate range (19) < Jam range (25) (more consistent).

Both claims are correct: Yes.

[4 marks] **[4]**

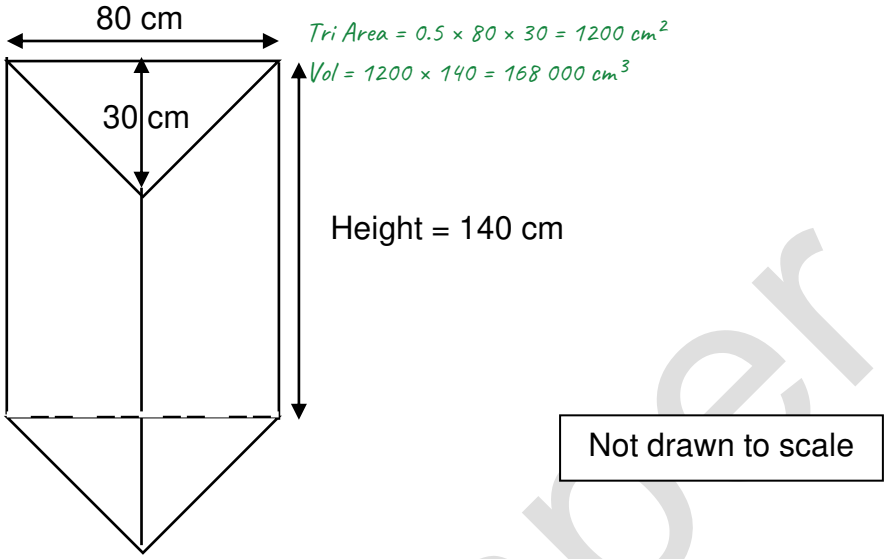
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14. Hayden needs to fill this fish tank with water.

The top of the tank is in the shape of a triangle.



Tri Area = 0.5 × 80 × 30 = 1200 cm²

Vol = 1200 × 140 = 168 000 cm³

To work out how much water the tank holds, Hayden uses this formula:

Volume of tank in cubic centimetres = area of top of tank × height of tank

Hayden has a bucket that will hold 11 litres (l) of water.

What is the minimum number of times Hayden will need to use the bucket to fill the tank?

Answer box is on the following page

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Step 1: Area of triangular top = 0.5 × base × height

Area = 0.5 × 80 cm × 30 cm = 1200 cm²

Step 2: Volume of tank = Area × height of tank:

Volume = 1200 cm² × 140 cm = 168 000 cm³

Step 3: Convert cm^3 to litres (1 litre = 1000 cm^3):

Capacity = 168 000 ÷ 1000 = 168 litres

Step 4: Number of 11-litre buckets = 168 ÷ 11 = 15.2727... times

Must round up to next full bucket: 16 times

Answer	16 ✓ [5 marks]

[5]

Past paper

Questions continue on the following page

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15. Luca wants to put ivy along the top of a 40-foot-long garden fence.

Luca finds these prices:

Ivy	A	B	C	D	E
Price per metre	£2.50	£2.00	£1.50	£1.00	£2.75

They will buy 18% more ivy than needed.

Luca buys the ivy that has the median price.

How much will Luca pay in total?

Use 1 foot = 0.305 metres

Show your working.

Step 1: Convert 40 feet to metres (1 ft = 0.305 m):

Length = 40 × 0.305 = 12.2 metres

Step 2: Find median price per metre:

Prices: £1.00, £1.50, £2.00, £2.50, £2.75 --> Median = £2.00/m

Step 3: Base cost = 12.2 m × £2.00/m = £24.40

Step 4: Add 18% extra ivy (multiply by 1.18):

Total = £24.40 × 1.18 = £28.792 --> Rounds to £28.79

Answer	£ 28.7 ✓ *[6 marks]*

[6]

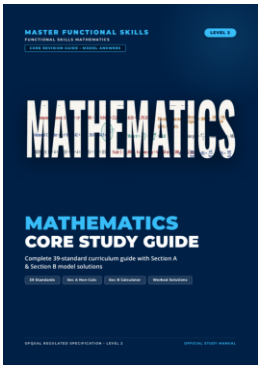
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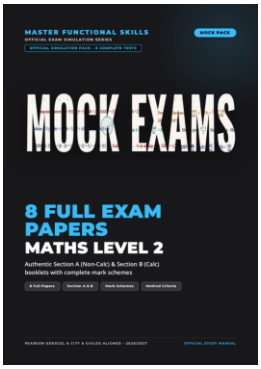


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